

Redocking Validation Reveals Structure-Dependent Docking Reliability in ATR: A Reproducible Pilot Virtual Screen of BRICS-Generated Candidates

[Author name(s) to be added]

[Affiliation to be added]

Correspondence: [email to be added]

Abstract

Background: Nearly all existing chemical probes of the ATR–CHK1 replication-stress checkpoint are ATP-competitive kinase inhibitors. Two near-atomic-resolution cryo-EM structures of the human ATR–ATRIP complex released in 2025 (PDB 9L40, bound to berzosertib/VE-822; PDB 9L4B, bound to camonsertib/RP-3500) provide new templates for structure-based work on this target, and 9L40 additionally revealed a second, allosteric berzosertib binding site at the ATR–ATR dimer interface distinct from the canonical ATP pocket. Whether newly deposited structures of this kind are equally reliable for conventional ATP-pocket docking is not something resolution alone can guarantee. **Methods:** We assembled a small seed library of five ATR/CHK1-directed compounds, three extracted directly from the co-crystallized active-site ligand of PDB 9L40, 9L4B, and a CHK1 reference structure (PDB 2YM8), generated derivatives by BRICS recombination, filtered for developability by standard descriptor rules, and – before docking any candidate – validated each receptor structure by redocking its own native ligand against a pre-specified 2.0 Å RMSD pass threshold, repeated across three independent seeded AutoDock Vina runs. **Results:** Redocking succeeded for 9L4B (mean RMSD 0.61 ± 0.02 Å) and 2YM8 (0.90 ± 0.01 Å) but reproducibly *failed* for 9L40 (2.96 ± 0.06 Å across all three seeds, despite tight search reproducibility of 0.56 Å pairwise), so 9L40 was excluded from candidate scoring per the pre-specified rule. A 40-compound pilot set (5 seeds + 35 developability-passed BRICS derivatives) was then docked against the validated 9L4B structure: predicted affinities ranged from -9.80 to -7.82 kcal/mol (best pose across three seeded replicates), with camonsertib itself – the receptor’s own co-crystallized ligand – ranking first as expected; no derivative exceeded the best seed compound, and the top-ranked derivatives were close BRICS analogs of camonsertib sharing its oxabicyclic-ether/pyrazolopyrazine core with a single substituent varied. **Conclusions:** A newly deposited, high-resolution structure is not automatically a reliable docking receptor, and this can only be established empirically, per structure, before any candidate is scored – not assumed from resolution or recency. All code, data, and structures used are openly available for independent verification.

Keywords: ATR kinase; molecular docking; redocking validation; BRICS; virtual screening; reproducibility; AutoDock Vina

Background

The ATR–CHK1 checkpoint kinase axis is the principal sensor of DNA replication stress: ATR is activated at stalled replication forks through protein-mediated mechanisms (the ATRIP–TOPBP1 axis, and independently ETAA1) rather than by occupation of its own ATP-binding pocket, and once active it phosphorylates and activates CHK1, which restrains further origin firing and protects replication forks from nucleolytic collapse (1–3). Genetic and pharmacological attenuation of ATR or CHK1 activity increases instability at common fragile sites – chromosomal loci prone to breakage under replication stress – consistent with this axis being protective (4, 5).

Essentially all existing chemical matter directed at this pathway – berzosertib, ceralasertib, camonsertib, prexasertib, and related compounds – is ATP-competitive kinase inhibitors developed for oncology synthetic-lethality strategies (6). Two cryo-EM structures of the full human ATR–ATRIP complex released in 2025, bound respectively to berzosertib (PDB 9L40) and camonsertib (PDB 9L4B) at ~ 3 Å resolution, are a substantial improvement over the original 2018 structure of the complex (PDB 5YZ0, 4.7 Å) and newly resolved that one ATR–ATRIP complex engages *four* berzosertib molecules – two in the canonical ATP-competitive active sites, and two at a previously uncharacterized allosteric pocket formed at the ATR–ATR dimer interface (7). This is a structurally more complex picture than a single ATP-pocket story, and it raises a practical question for anyone using these new structures for docking: does a newly deposited, high-resolution structure automatically make a trustworthy receptor for conventional ATP-pocket virtual screening?

This is an empirical question, not one that follows from resolution or recency, and it is the question this pilot study is organized around. Before screening any candidate against 9L40 or 9L4B, we validate each structure’s suitability as a docking receptor by redocking its own co-crystallized ligand – the standard control for docking-protocol accuracy (8) – and only proceed to score novel candidates against a structure that passes. We then report a modest pilot screen of BRICS-derived (9) candidates against whichever structure(s) pass, as a demonstration of the validate-before-screening principle applied to a concrete, currently topical target.

Scope

This is explicitly a pilot study, not a comprehensive virtual screening campaign. Docking, by itself, estimates predicted binding and cannot establish whether a ligand activates, inhibits, or otherwise modulates ATR or CHK1 – physiological ATR activation depends on upstream protein interactions absent from a static structural model (1, 2). All results below are reported as predicted binding and prioritization, not as demonstrated pharmacology. Molecular dynamics and MM-GBSA relative energetic estimation, and a broader cross-kinase selectivity panel, were part of the original design of the computational pipeline this pilot draws on, but are not included here: they require substantially more compute (GPU-accelerated MD) and additional validated receptor structures than were available for this pilot, and are explicitly left as future work (see Limitations) rather than run at a scale too small to be informative.

Methods

Receptor structures

Three receptor structures were obtained, in mmCIF format, via the Registry of Open Data on AWS mirror of the RCSB PDB archive ([s3://pdbsnapshots](https://pdbsnapshots.s3.amazonaws.com/), snapshot 2026-01-01), since the primary RCSB PDB web/FTP endpoints were not reachable from the compute environment used for this study: PDB 9L40 (ATR kinase domain bound to VE-822/berzosertib, ~ 2.9 Å), PDB 9L4B (ATR kinase domain bound to camonsertib/RP-3500, ~ 3 Å), and PDB 2YM8 (human CHK1 kinase domain bound to a chloroisoquinolinyllamino-pyrazine-carbonitrile inhibitor, 2.07 Å) (7). The original 2018 ATR–ATRIP structure (PDB 5YZ0, 4.7 Å overall) was not used as a docking receptor: at that resolution, ATP-pocket side-chain positions are not reliably resolved.

For each structure, protein chains were extracted (standard residues only, via Biopython (10)), and the co-crystallized ligand of the biologically relevant ATP-competitive active site was identified and isolated. For 9L40, four copies of the ligand (chemical component A1E1E) are present per asymmetric unit; contact analysis (heavy atoms within 4.5 Å of each protein chain) confirmed that residues A2701/B2701 each contact only their own chain (canonical, single-monomer ATP-competitive site)

while A2702/B2702 contact both chains (the dimer-interface allosteric site reported by Wang et al. (7)); A2701 was used as the active-site reference ligand. The analogous single-copy active-site ligands (chemical component A1EIK for 9L4B; YM8 for 2YM8) were extracted directly. New-style five-character PDB chemical-component codes overflow the fixed-width residue-name column of the legacy PDB format and silently corrupt naive PDB-to-PDBQT conversion (each atom is parsed as an isolated one-atom fragment); ligand records were therefore rewritten with a placeholder three-character residue name before conversion. Proteins and ligands were converted to PDBQT with Open Babel (11), and docking search boxes were centered on each active-site ligand’s heavy-atom centroid (22 Å cubic box).

Seed compound library

Five compounds formed the seed library (Table 1): the three active-site ligands extracted directly from the co-crystal structures above (berzosertib, camonsertib, and the 2YM8 CHK1 inhibitor), plus prexasertib and ceralasertib, whose canonical SMILES were obtained via literature search and independently cross-checked against reported molecular formula/exact mass (both matched exactly: prexasertib $C_{18}H_{19}N_7O_2$, exact mass 365.16; ceralasertib $C_{20}H_{24}N_6O_2S$). The PubChem and ChEMBL REST APIs used by this pipeline’s programmatic seed-retrieval script were not reachable from the compute environment used for this study, so this smaller, directly-verified seed set was used in their place; all five structures were confirmed valid and standardized with RDKit (12) (salt stripping, charge neutralization, canonicalization).

Table 1: Seed compound library. Compounds marked * were extracted directly from the co-crystallized active-site ligand of the named PDB structure; the remaining SMILES were obtained via literature search and independently verified against reported molecular formula and exact mass.

Compound	Target	Provenance
Berzosertib (VE-822)	ATR	PDB 9L40, active-site ligand*
Camonsertib (RP-3500)	ATR	PDB 9L4B, active-site ligand*
CHK1 inhibitor (YM8)	CHK1	PDB 2YM8, active-site ligand*
Prexasertib	CHK1	Literature SMILES, formula-verified
Ceralasertib (AZD6738)	ATR	Literature SMILES, formula-verified

Derivative generation and developability filtering

The seed set was fragmented with the BRICS algorithm (9) into 22 unique fragments, recombined (BRICS.BRICSBUILD, RDKit), and capped at 5,000 generated candidates (a combinatorial cap, not a natural stopping point – see Limitations). Candidates were pre-filtered by a relaxed Lipinski rule (≤ 1 violation of molecular weight ≤ 500 , cLogP ≤ 5 , H-bond donors ≤ 5 , H-bond acceptors ≤ 10) (13), Veber’s rule (rotatable bonds ≤ 10 , TPSA $\leq 140 \text{ Å}^2$) (14), and removal of any structure matching the RDKit PAINS (15) or Brenk structural-alert catalogs, leaving 786 candidates. Every retained candidate (plus the 5 seeds) was then scored against four descriptor-based drug-likeness rules (Lipinski, Veber, Ghose (16), Egan (17)); 680 of 791 compounds satisfied $\geq 3/4$ rules with no structural alert (“developability-passed”). All values from this stage are rule-based predictions, not measurements.

Redocking validation

Before any candidate was scored, each receptor was validated by extracting its own active-site ligand, redocking it into the identical receptor/box with AutoDock Vina (8) (exhaustiveness 32, three independent seeded replicates: 1000, 1001, 1002), and computing the root-mean-square deviation (RMSD, heavy

atoms, index-matched – Vina’s PDBQT output preserves input atom order) between the best-scoring pose and the native crystallographic pose, against a pre-specified pass threshold of ≤ 2.0 Å, fixed before any receptor was tested. A receptor failing this control in any replicate is excluded from candidate scoring, with the reason stated, rather than retried until it passes. Search reproducibility (pairwise RMSD between replicate best poses, independent of whether that pose is correct) was reported separately from redocking accuracy (RMSD to the native pose), since these answer different questions.

Pilot library docking

The 5 seed compounds plus the top 35 developability-passed derivatives (ranked by number of rules satisfied, then by molecular weight ascending) were docked against every receptor structure that passed redocking validation, using the identical protocol (AutoDock Vina, exhaustiveness 16, three independent seeded replicates: 2000, 2001, 2002). Ligands were 3D-embedded and protonated at pH 7.4 with Open Babel. The best (most negative) affinity across replicates is reported per compound, alongside the mean and standard deviation across replicates.

Computational environment

All calculations were run with AutoDock Vina 1.2.7 (Python bindings) and Open Babel (both via pip, CPU only, no GPU available in this environment – see Limitations regarding molecular dynamics), RD-Kit 2026.03.6, and Biopython 1.88. Receptor structures were retrieved from the RCSB PDB AWS Open Data mirror snapshot dated 2026-01-01. All code and intermediate data are available at the repository accompanying this submission (see Availability of data and materials).

Results

Redocking validation: one of two ATR structures is unreliable for docking

Table 2 summarizes the redocking validation across all three receptor structures over three independent seeded replicates. 9L4B (ATR, camonsertib-bound) and 2YM8 (CHK1) both passed with a substantial margin (mean RMSD 0.61 ± 0.02 Å and 0.90 ± 0.01 Å respectively, well under the 2.0 Å threshold in every replicate). 9L40 (ATR, berzosertib-bound) failed reproducibly: mean RMSD 2.96 ± 0.06 Å, exceeding the threshold in all three replicates. This was not simply search noise – the pairwise RMSD between 9L40’s own replicate best poses (search reproducibility) was tight (0.56 Å mean), meaning Vina consistently converged on the same, wrong, pose under this box/protocol, rather than failing to converge at all. Per the pre-specified rule (Methods), 9L40 was excluded from candidate scoring; the pilot library docking below uses 9L4B only.

Table 2: Redocking validation across three independent seeded Vina replicates (exhaustiveness 32).

Structure	Mean RMSD (Å)	Passes ≤ 2.0 Å	Search reproducibility (Å)
9L40 (ATR, VE-822)	2.96 ± 0.06	No (all 3 replicates)	0.56
9L4B (ATR, camonsertib)	0.61 ± 0.02	Yes (all 3 replicates)	0.045
2YM8 (CHK1)	0.90 ± 0.01	Yes (all 3 replicates)	0.078

Seed and derivative library

The 5-compound seed set yielded 22 unique BRICS fragments, from which 5,000 recombination candidates were generated (capped) and 786 passed the initial Lipinski/Veber/PAINS/Brenk pre-filter. Of the resulting 791 compounds (786 derivatives + 5 seeds), 680 (86%) satisfied $\geq 3/4$ developability rules with no PAINS/Brenk structural alert.

Pilot docking against the validated ATR structure

The 40-compound pilot set (5 seeds + 35 developability-passed derivatives) was docked against 9L4B (exhaustiveness 16, three seeded replicates each). Best-pose predicted affinity across the set ranged from -9.80 to -7.82 kcal/mol (mean -8.66 ± 0.45 kcal/mol across all 40 compounds). Table 3 shows the top 10 ranked compounds. Camonsertib – the receptor’s own co-crystallized ligand – ranked first, as expected for a structure the receptor was solved with. No BRICS derivative exceeded the best seed compound’s predicted affinity; the closest, deriv_002 (-9.47 kcal/mol), came within 0.33 kcal/mol of camonsertib. The three top-ranked derivatives (deriv_002, deriv_003, deriv_019) share camonsertib’s oxabicyclic-ether/pyrazolopyrazine core with a single substituent varied (methylamino, methoxy, and dimethylamino respectively, in place of camonsertib’s own morpholine substituent) – an interpretable and expected outcome of BRICS recombination re-using the best-scoring seed compound’s own fragments in a new context, rather than an independent discovery of novel chemical matter.

Table 3: Top 10 ranked compounds from the 40-compound pilot docking against 9L4B (mean $\pm$ SD and best across 3 seeded replicates, exhaustiveness 16).

Compound	Source	Mean affinity (kcal/mol)	Best affinity (kcal/mol)
camonsertib (seed)	seed	-9.78 ± 0.01	-9.80
deriv_002	derivative	-9.46 ± 0.01	-9.47
ceralasertib (seed)	seed	-9.39 ± 0.01	-9.40
deriv_003	derivative	-9.33 ± 0.01	-9.34
deriv_019	derivative	-9.25 ± 0.03	-9.29
deriv_032	derivative	-9.06 ± 0.17	-9.15
deriv_022	derivative	-9.08 ± 0.02	-9.10
berzosertib (seed)	seed	-8.94 ± 0.05	-8.99
deriv_033	derivative	-8.96 ± 0.02	-8.98
deriv_026	derivative	-8.94 ± 0.00	-8.94

Discussion

The central finding of this pilot is methodological rather than chemical: **a newly deposited, higher-resolution structure is not automatically a reliable docking receptor, and only an explicit redocking control establishes this, per structure, before any candidate is trusted.** 9L40 and 9L4B were solved in the same study, at comparable resolution (7), yet only one reproducibly passed a standard accuracy control under an otherwise identical protocol. A screening campaign that skipped this control and used 9L40 by default (e.g., because it was the first hit in a search, or because it captures the more novel allosteric site) would have scored every candidate against a receptor conformation the docking protocol itself cannot reliably reproduce. Whether the 9L40 failure reflects a genuine limitation of rigid-receptor Vina docking against this particular conformational state (e.g., an induced-fit effect specific to the dimer-interface-engaged assembly) or a more mundane box/protocol mismatch is not resolved by this pilot and

is worth follow-up, but the practical implication for anyone reusing these structures is the same either way: validate before you screen.

The pilot docking result is, appropriately for a 35-derivative pilot, modest rather than dramatic: no generated derivative exceeded the predicted affinity of the seed compounds, and the best-performing derivatives are recognizable as close analogs of camonsertib rather than structurally novel chemotypes. This is consistent with a BRICS recombination run over a 5-compound seed set and a fragment pool of only 22 pieces: the search space explored is inherently local to the seed chemistry, and a pilot at this scale should not be expected to surface a structurally distinct chemotype with superior predicted affinity. What this pilot does show is that the pipeline behaves sensibly at small scale – it recovers the seed compound’s own high rank (unsurprising, since camonsertib is the receptor’s native ligand, but a useful sanity check that the docking protocol is not producing arbitrary rankings) and produces derivatives whose structure-score relationship is interpretable (small, defined substituent changes at one position, with correspondingly small score changes) rather than erratic. Whether a larger seed set, a larger fragment pool, or multiple recombination rounds would surface derivatives that improve on the seed compounds is an open question this pilot’s scale cannot answer.

Limitations

This is a small pilot, and its scope was set by what could be executed directly and verifiably, not by what would be most impressive to report. Specifically: (1) the BRICS recombination cap (5,000) is combinatorial, not principled, and a larger or differently seeded run would sample a different, likely larger, region of the accessible chemical space; (2) the 40-compound pilot library is a small, deterministic subset of the developability-passed derivatives (ranked by rule count then molecular weight), not a comprehensive screen, and enrichment-style retrieval statistics (ROC-AUC, enrichment factor) are not meaningful at this scale and are not reported; (3) no molecular dynamics or MM-GBSA relative energetic estimation was run – both require substantially more compute (GPU acceleration for MD at any informative timescale) than was available, and docking scores alone are known to correlate imperfectly with binding affinity; (4) no cross-kinase selectivity panel was run – this would require additional receptor structures, each independently redocking-validated by the same standard applied to the three structures here, which was beyond this pilot’s scope; (5) as throughout, nothing here establishes ATR/CHK1 functional modulation, inhibition, or activation – only predicted binding and ranking; (6) the seed and reference SMILES for prexasertib and ceralasertib were obtained via web search rather than a structured chemical database query (PubChem/ChEMBL were not reachable) and, while independently verified against reported molecular formula and exact mass, should be re-confirmed against a primary chemical database before further use.

Conclusions

Redocking validation – not resolution, recency, or biological interest – is what determines whether a deposited structure is a trustworthy docking receptor. Applied to two contemporaneous, comparable-resolution ATR structures, this control passed one and failed the other, a distinction that would not have been visible without running it. Screened against the validated structure, a small BRICS-derived pilot library reproduced the expected top rank of the receptor’s own native ligand and generated interpretable, if modest, analogs of it, without yet surfacing a chemotype that improves on the seed compounds – a realistic, honestly reported outcome for a pilot of this scale, and one further reason to treat larger claims from similarly small screens with caution. All structures, code, and intermediate data used in this study are openly available for independent verification and reuse.

Availability of data and materials

All code, the seed and derivative compound tables, redocking validation results, and pilot docking results are available at <https://github.com/ethanyyy123/research-molecule-editing> (branch [claude/cadd-fragile-sites-optimization-0u9fsp](#) at the time of this submission; an archived, versioned release should be minted, e.g. via Zenodo, before final submission). Receptor structures are publicly available from RCSB PDB (9L40, 9L4B, 2YM8) and were retrieved via the AWS Open Data mirror ([s3://pdbsnapshots](#)).

Competing interests

[PLACEHOLDER – e.g. “The authors declare no competing interests,” if accurate; confirm before submission.]

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Authors’ contributions

[PLACEHOLDER – e.g. following CRediT roles: conceptualization, methodology, software, formal analysis, investigation, writing.]

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